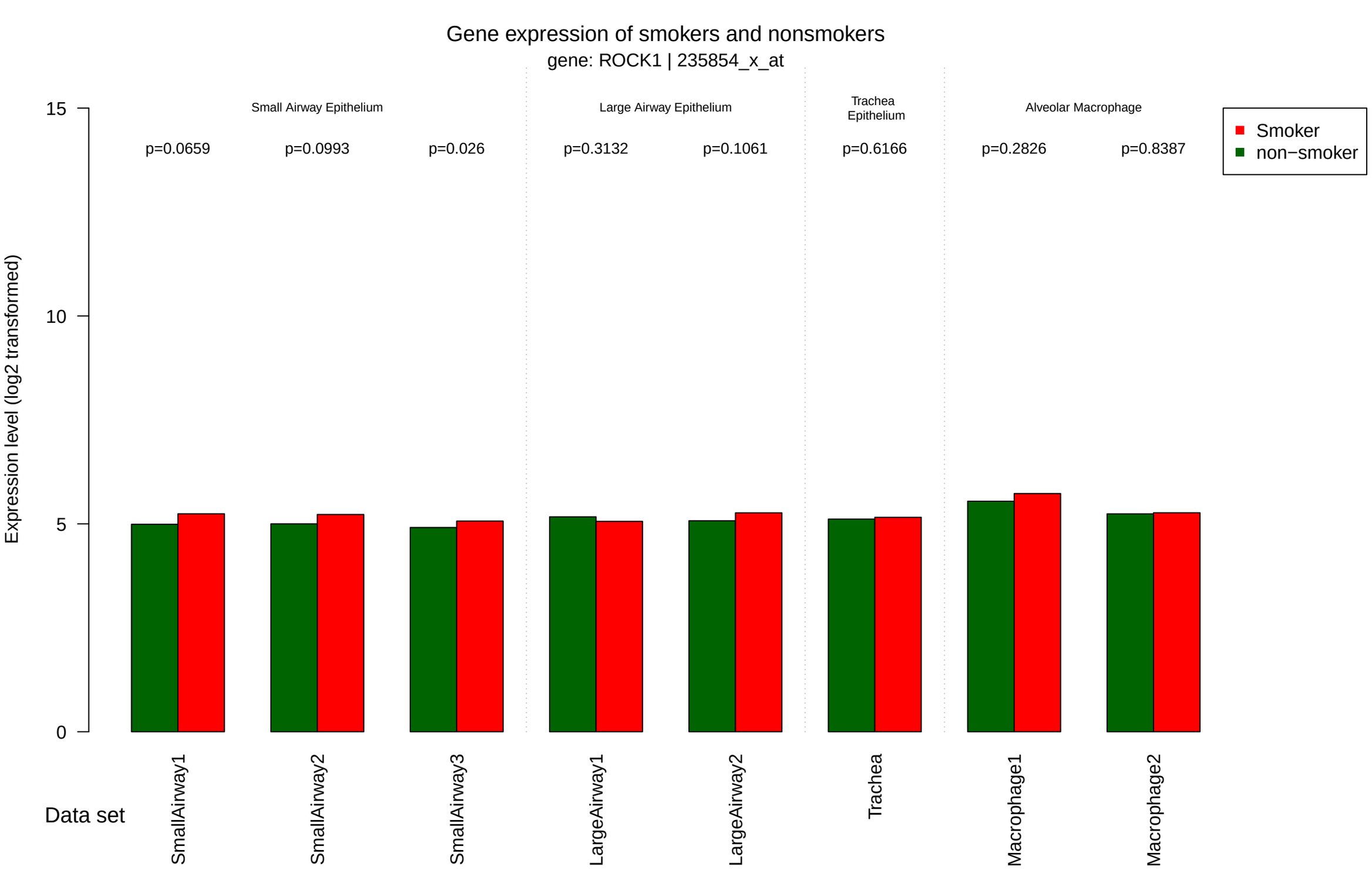
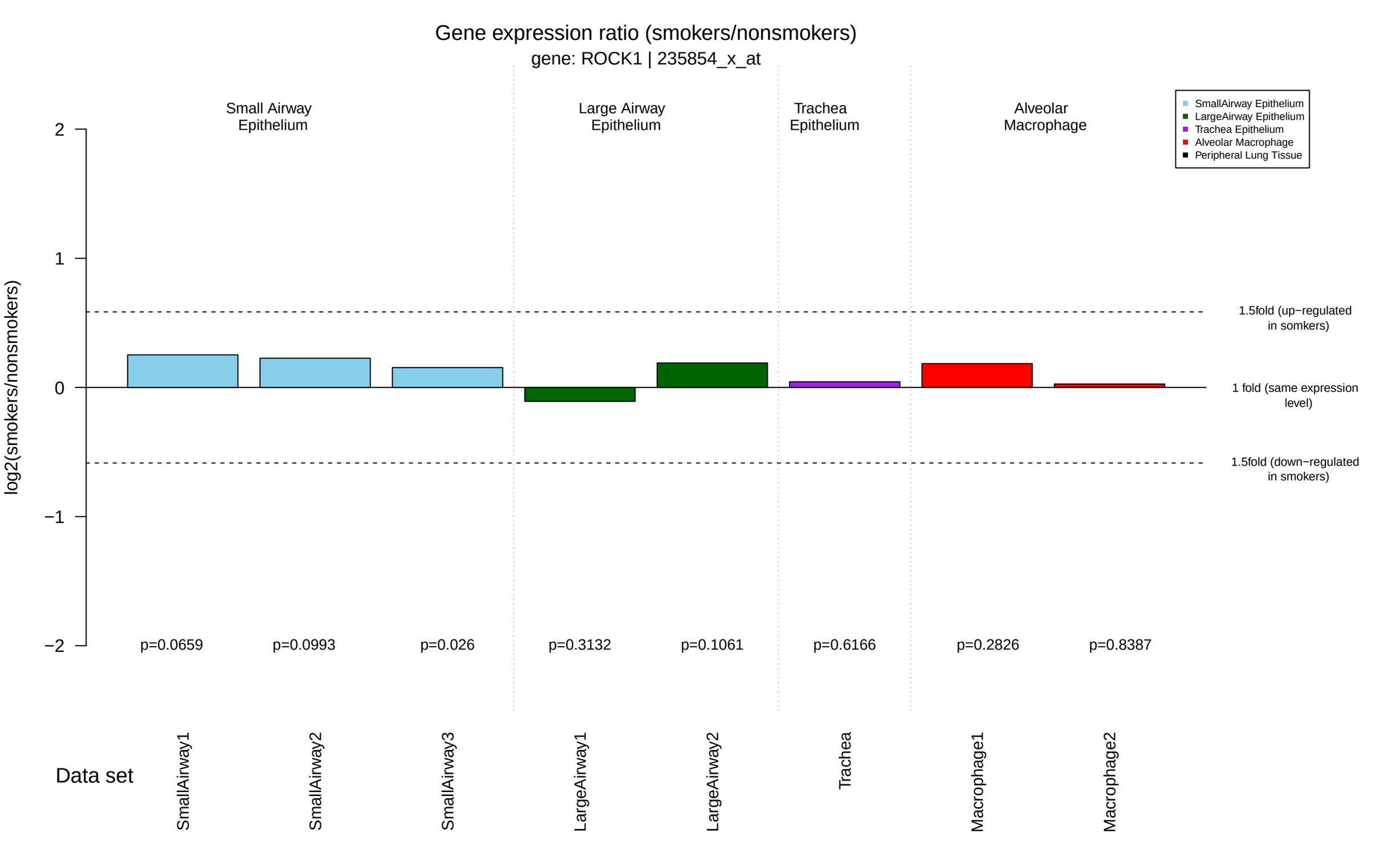
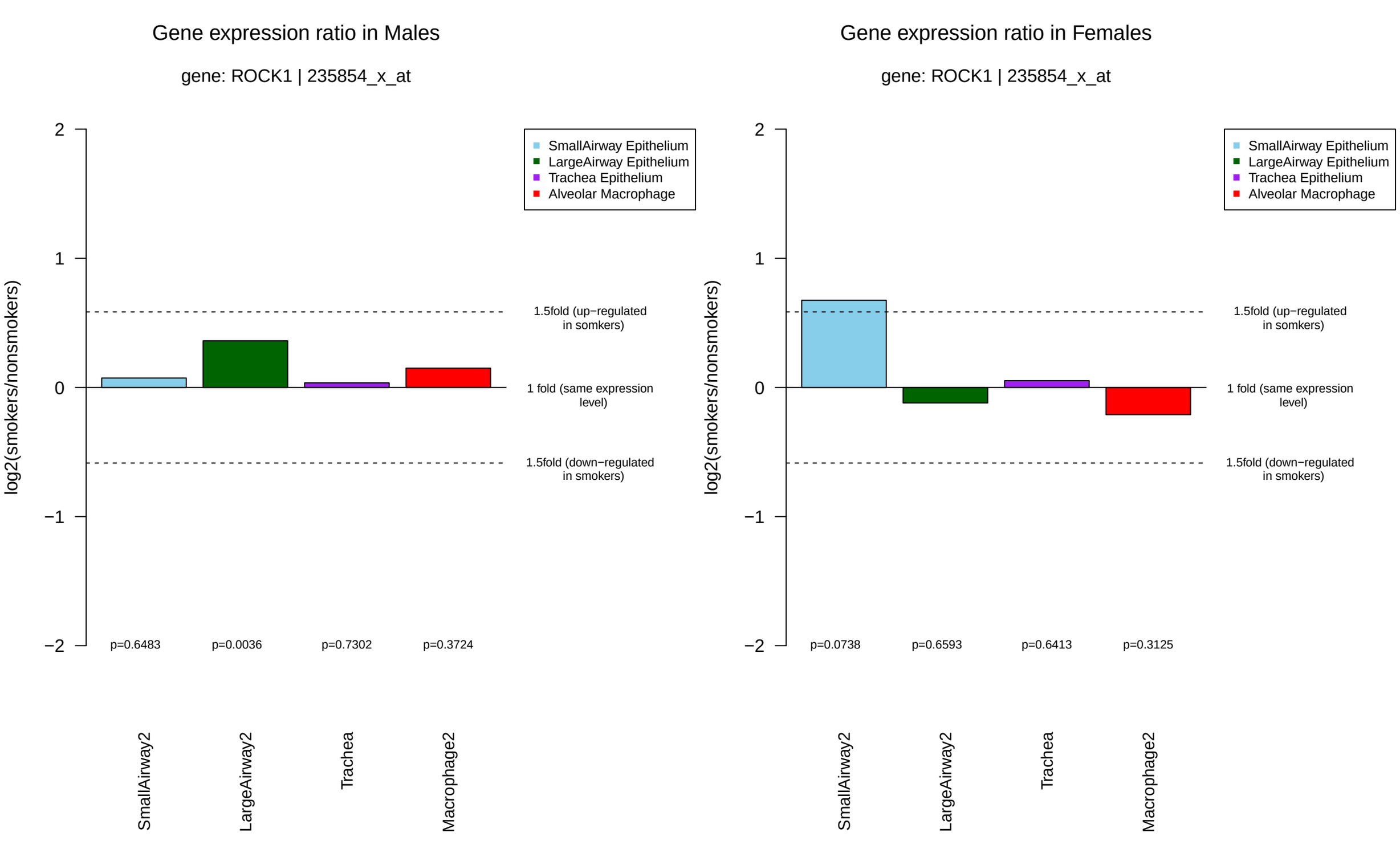
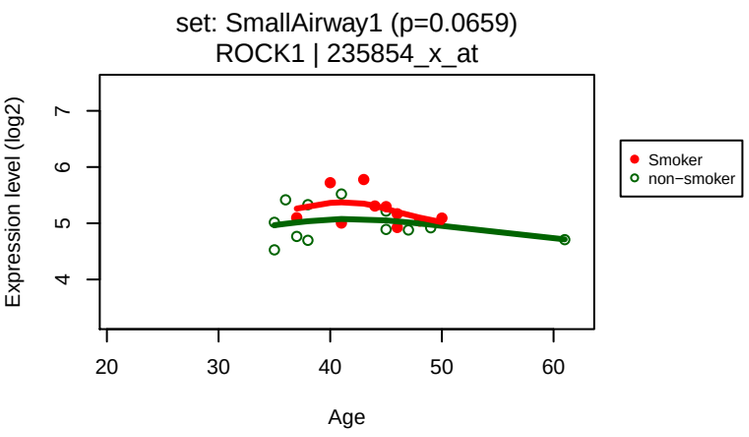
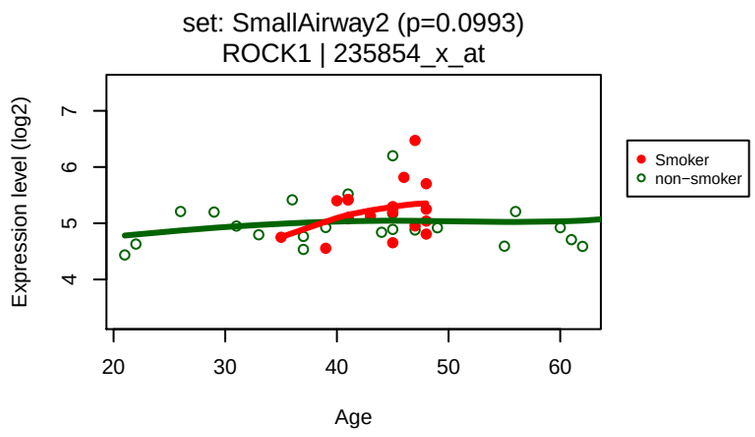
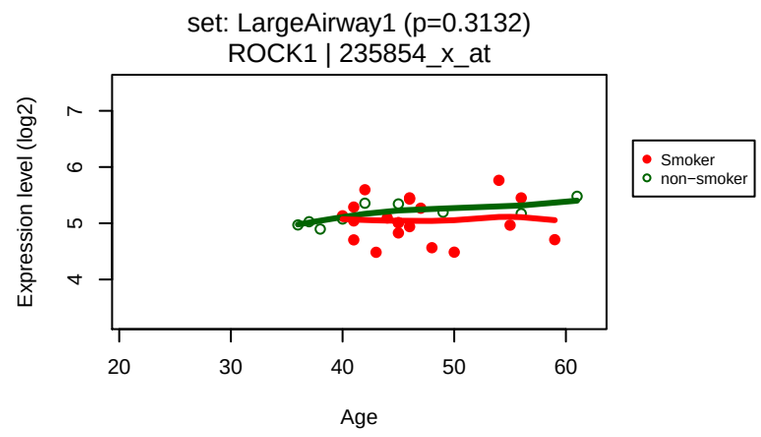

Gene expression of smokers and nonsmokers
gene: ROCK1 | 235854_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0659
p=0.0993
p=0.026
p=0.3132
p=0.1061
p=0.6166
p=0.2826
p=0.8387
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

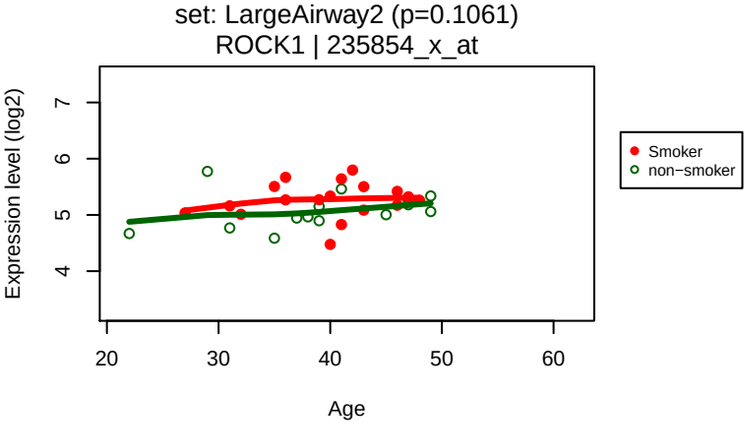

Gene expression ratio (smokers/nonsmokers)
gene: ROCK1 | 235854_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0659
p=0.0993
p=0.026
p=0.3132
p=0.1061
p=0.6166
p=0.2826
p=0.8387
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

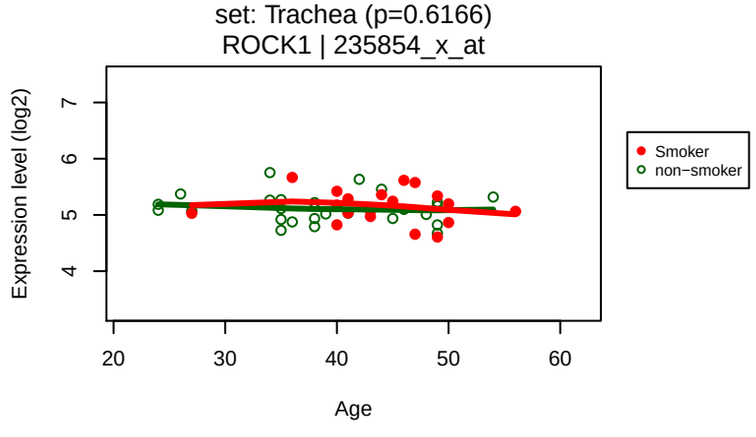

Gene expression ratio in Males
gene: ROCK1 | 235854_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6483
p=0.0036
p=0.7302
p=0.3724
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ROCK1 | 235854_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0738
p=0.6593
p=0.6413
p=0.3125
SmallAirway2
LargeAirway2
Trachea
Macrophage2

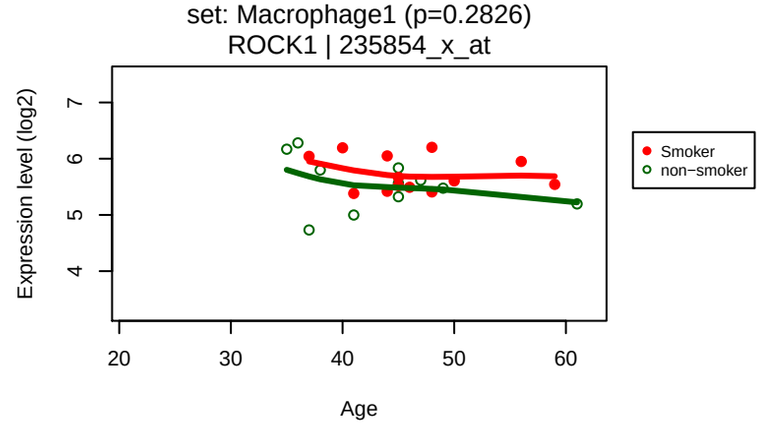

set: SmallAirway1 (p=0.0659)
ROCK1 | 235854_x_at
Expression level (log2)
Age
Smoker
non−smoker

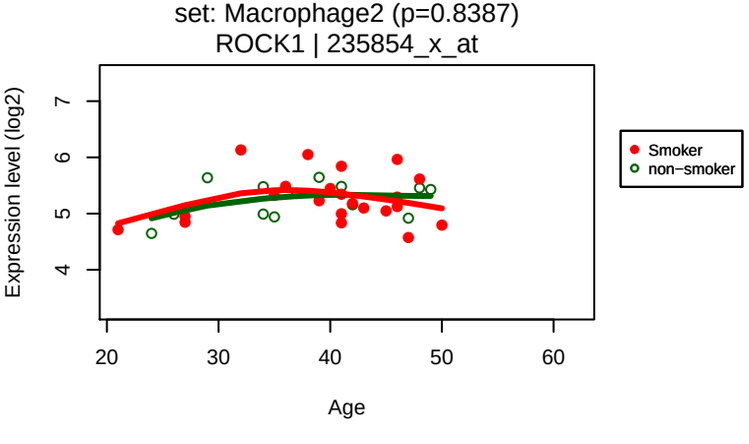

set: SmallAirway2 (p=0.0993)
ROCK1 | 235854_x_at
Expression level (log2)
Age
Smoker
non−smoker

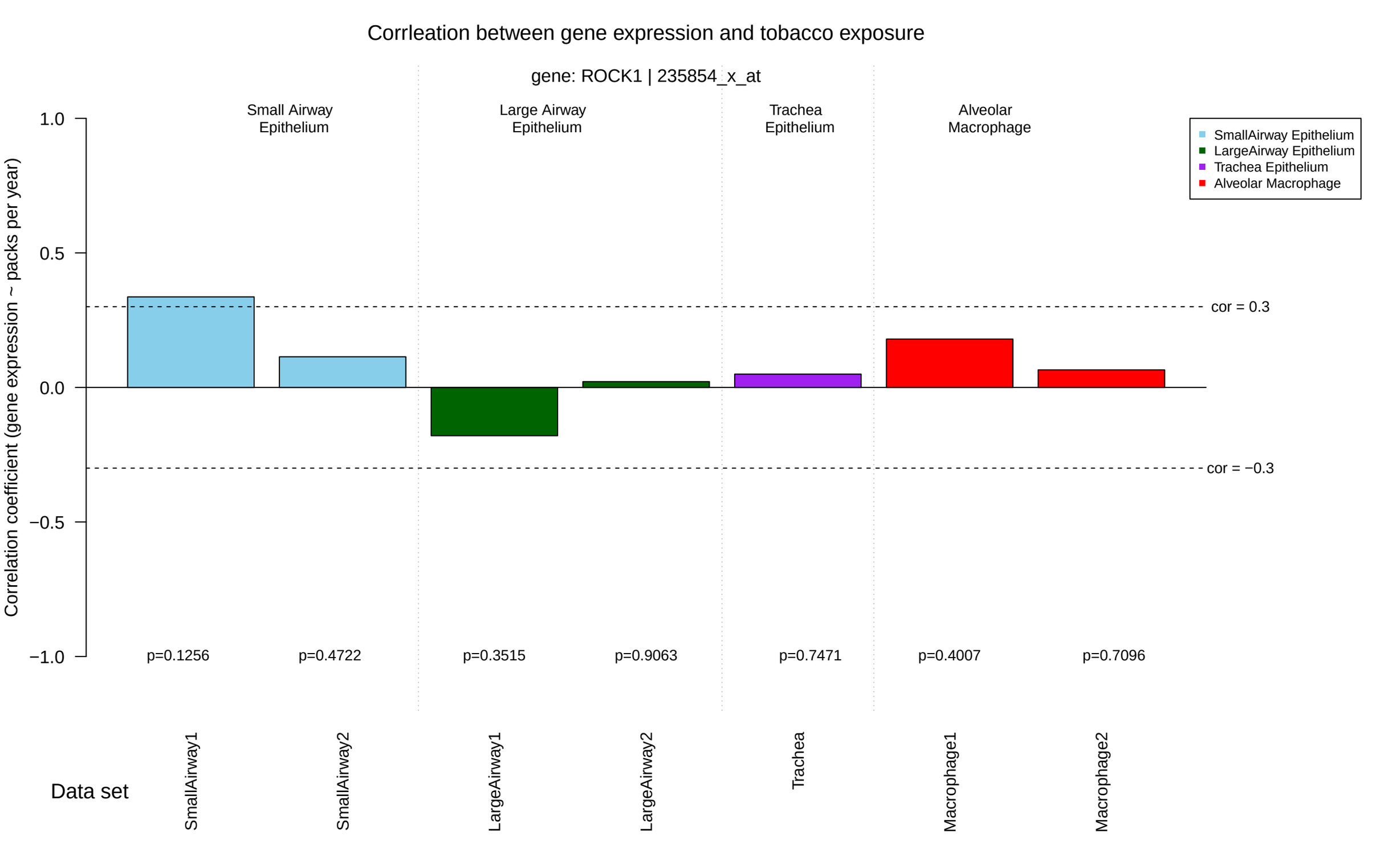

set: LargeAirway1 (p=0.3132)
ROCK1 | 235854_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1061)
ROCK1 | 235854_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6166)
ROCK1 | 235854_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2826)
ROCK1 | 235854_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8387)
ROCK1 | 235854_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ROCK1 | 235854_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1256
p=0.4722
p=0.3515
p=0.9063
p=0.7471
p=0.4007
p=0.7096
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2